STAT 251 Project Specifications

Project Proposal Due October 30th, 2026
Final Project Due December 9th, 2026

Overview

You will complete this project with a partner. Together, you will pose a research question, find data that can inform it, decide on a statistical model for the data, specify and defend priors for the model parameters, determine/approximate the posterior distributions, and use those distributions to make inference and conclusions about your research question.

Originality and AI. The analysis that you do should be original and represent your own ideas. Do not copy another analysis you find, or farm out the creative process to AI. In fact, the only thing I would like you to use AI for is for help with beautifying your plots in R and suggesting copy edits (JUST COPY EDITS) to your write-ups. This applies to both the project proposal and the final project write-up.

Project Proposal (Due October 30th, 2026)

Submit one document per group containing:

- The names of both partners.
- Your research question(s). You should have one principal research question in mind, but you may have another question or two if they are relevant to the main question. Importantly, your research question must involve a comparison of two or more populations, groups, or treatments. You are welcome (but not required) to frame your research question around a null and alternative hypothesis.
- The data source and a short written description of the data set (including a URL if possible). If the data are not publicly available, explain where they come from and how you will share them with me.
- A well-organized and well-articulated exploratory analysis of the data, including well-labeled plots created in R. Make sure to explain in prose any plots and/or statistics in your report. This exploratory analysis should be connected to your research question—don't include anything unrelated to the research question.

If you do your proposal in Quarto, do not echo your code chunks or print raw R output (with the exception of plots). Any R output should be properly formatted within the text itself or as well-formatted tables (see the kable package).

I will use your proposal to give you feedback and, if necessary, redirection. Please come chat with me about your ideas and any questions that you have! I can be a great help to you!

The project proposal will be worth 20% of the total point value of the project.

Final Report (Due December 9th, 2026)

Write your final report for someone who understands Bayesian statistics but is not an expert in the subject area of your research question. Explain any background or terminology a reader would

need to follow your analysis, but you do not need to explain Bayesian methods from scratch.

If you do your proposal in Quarto, do not echo your code chunks or print raw R output (with the exception of plots). Any R output should be properly formatted within the text itself or as well-formatted tables (see the kable package). You will include your R code in an appendix at the end of the project (see below).

Your final writeup must include the following sections:

- **Introduction:** Explain your research question, relevant background information about it, and why it matters to you.
- **Data:** Explain how the data were obtained, and cite their source. Summarize the data with relevant summary statistics and figures.
- **Methods:** Identify the observational unit, select a data model (likelihood), and justify this choice (i.e., determine whether its assumptions are reasonable, and if not, describe some possible consequences). Explain what the parameters of the likelihood are and how they will help you answer your question. Choose and defend your prior distributions.
- **Results:** Perform posterior inference to answer your research question. Your analysis must compare two or more populations through a function of their parameters (for example, a difference in means, $\mu_1 - \mu_2$, or a ratio of variances, σ_1^2/σ_2^2). Report and properly interpret a credible interval for this function, and include well-formatted and informative graphic(s) of its posterior distribution together with its corresponding prior. Where appropriate, also report and interpret posterior probabilities of hypotheses relevant to your question (e.g., $P(\mu_1 > \mu_2 \mid \mathbf{y})$).
- **Discussion:** Summarize your results, identify shortcomings of the work, and explain what could be done in the future to better answer your question or to address a natural follow-up question.
- **Appendix:** Include all R code necessary to reproduce your analysis.

Ideally, the Introduction through Discussion will be about 1–1.5 pages long, single spaced (not including graphs).

Please reach out to me and the TAs if questions arise.